

# On the cyclic encoding of control<sub>2</sub>

Semantic Control Systems (1995), Definition 28

Claude (Fable 5.1), as prompted by Shingai Thornton\*  
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**Source.** C. Joslyn, “Semantic Control Systems”, *World Futures* 45 (1995) 87–123, cited below as J95 with the journal’s page numbers. Definition 28 (p. 103): “A control<sub>1</sub> system  $CS = \langle C, O \rangle$  is also a control<sub>2</sub> system if the variation in the constraint<sub>1</sub> on  $O$ , induced by  $C$ ’s variation, is itself constrained<sub>2</sub>.” Proposition 29 (p. 106) splits the object of control,  $O = \langle O_E, O_I \rangle$ , giving

$$CS = \langle C, O = \langle O_E, O_I \rangle \rangle \quad (30)$$

and §3.5 (p. 107) names the relations among the three components (Figure 1): “Thus together  $O_E$  and  $O_I$  form an afferent-efferent loop.”

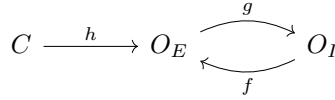
This note records the three named relations of Figure 1 as the arrows of a quiver and takes the free category on it (§1); a theorem follows from the loop alone (§2); §3 says what it does not claim; §4 asks whether the drawing is faithful to what you meant; §5 is a separate provenance item.

## 1 The encoding

**Definition 1** (the quiver  $Q_J$ ). *Vertices*  $\{C, O_E, O_I\}$ . *Generating arrows*

$$h: C \rightarrow O_E, \quad g: O_E \rightarrow O_I, \quad f: O_I \rightarrow O_E.$$

Convention. *Arrows point in the functional direction of Figure 1* (p. 107),  $x \rightarrow y$  meaning “ $x$  acts on  $y$ ”:  $C$  disturbs  $O_E$ ;  $O_E$  constrains  $O_I$ ;  $O_I$  feeds back to  $O_E$ .



| symbol | role in $Q_J$ | J95 (page)                                                                                                   |
|--------|---------------|--------------------------------------------------------------------------------------------------------------|
| $C$    | vertex        | “the global environment” (107); “a source of variation to the system $O$ ” (107)                             |
| $O_E$  | vertex        | “some efferent component of $O$ , the mechanism whose actions assert compensating variation on $O_I$ ” (107) |
| $O_I$  | vertex        | “the controlled variables” (107)                                                                             |
| $h$    | arrow         | “The variation of $C$ affects $O$ through the relation $h$ ” (106)                                           |
| $g$    | arrow         | “the final constraint <sub>1</sub> of $O_E$ on $O_I$ ” (106)                                                 |
| $f$    | arrow         | “whatever relation there might be back from $O_I$ to $O_E$ ” (106)                                           |

**Definition 2** (the shape).  $\mathcal{I}_J := \mathbf{Paths}(Q_J)$ , the free category on  $Q_J$ : objects are the vertices, morphisms are finite directed paths, composition is concatenation, identities are empty paths. No relations are imposed.<sup>1</sup>

*Remark 1* (what a quiver forgets).  $Q_J$  records only that a named dependency exists. It does not carry: (a) that  $f, g, h$  are functions rather than relations (§4.3, p. 114); (b) the asymmetry of the loop (§4.3, p. 114:  $f$  and  $g$  “are not thereby symmetric or complementary”); (c) the hierarchy of  $C$  over  $O$  and the “large loop gain” (p. 108); (d) the second loop. §3.7.2 (p. 110) says “there is also a kind of loop between  $C$  and  $O_E$ ” by reciprocal variation, and §3.7.1 (p. 109) rules it one-way at the level of action (“ $O_E$  mirror the actions of  $C$ , but not vice versa”).  $Q_J$  follows §3.7.1:  $h$  is a single arrow. (e) Turchin’s alternative, in which  $g$  “flows only through the environment” (p. 108). Items (d) and (e) are encoding forks, not oversights; each yields a different  $Q$ .

## 2 The theorem

Write  $\ell := f \circ g \in \text{Hom}_{\mathcal{I}_J}(O_E, O_E)$  for the feedback loop, a path of length 2.<sup>2</sup>

**Lemma 1.** *For  $n \in \mathbb{N}$ ,  $\ell^n$  has length  $2n$ ; hence  $n \mapsto \ell^n$  is injective and  $\text{Hom}_{\mathcal{I}_J}(O_E, O_E)$  is infinite.<sup>3</sup>*

*Proof.* Length is additive under concatenation, so  $|\ell^n| = n|\ell| = 2n$ . Distinct  $n$  give distinct lengths, hence distinct paths.  $\square$

**Theorem 1.** *Let  $\mathcal{C}$  be any category in which every hom-set is finite. No functor  $F: \mathcal{I}_J \rightarrow \mathcal{C}$  is faithful.<sup>4</sup>*

\*Byline per the Multivar repository’s convention. The encoding decision, the readings of J95, and §4 are Shingai Thornton’s; §2 is machine-checked in [github.com/halcyonic-systems/systems-science-foundations](https://github.com/halcyonic-systems/systems-science-foundations), files `Systems/Category/{ShapeJoslyn, CyclicObstruction, JoslynIncomparability}.lean`, commit 4288793.

<sup>1</sup>`JoslynShape := Paths JoslynPosition, ShapeJoslyn.lean`. `Paths` and `Path.length.comp` are Mathlib’s.

<sup>2</sup>`joslynLoop, joslynLoop.length : joslynLoop.length = 2 := rfl`.

<sup>3</sup>`cyclePow.length, cyclePow.injective, infinite_path_of_cycle`, all proved for an arbitrary quiver and an arbitrary positive-length loop.

<sup>4</sup>`joslyn.no_faithful_functor`, instance of the general `paths.no_faithful_functor_of_cycle`. Axioms: `propext`, `Classical.choice`, `Quot.sound`. No sorry.

*Proof.* A faithful functor is injective on each hom-set. By Lemma 1 it would inject the infinite set  $\text{Hom}(O_E, O_E)$  into the finite set  $\text{Hom}(FO_E, FO_E)$ .  $\square$

**Corollary 1.** *Let  $\mathcal{I}_K := \mathbf{Paths}(R \rightarrow T)$  be the shape of Klir’s  $S = (T, R)$ : two objects, one generating arrow. Every hom-set of  $\mathcal{I}_K$  is a subsingleton, hence finite. There is no faithful functor  $\mathcal{I}_J \rightarrow \mathcal{I}_K$ .<sup>5</sup>*

*Remark 2* (why this is the boundary and not a curiosity). Every other tradition encoded in the same program (Klir, Bunge, Mobus, Myers, Wymore, Mesarović) has an acyclic  $Q$ , hence finite hom-sets, and is placed by faithful comparison functors;  $Q_J$  is the first cyclic shape (Spivak’s value loop is the other). Feedback is where the finite-shape method stops, and Theorem 1’s sole premise is the cycle.

### 3 What the theorem does not say

1. **It is presentation-relative.** From the theorem’s own docstring, written before anyone asked you: “both shapes are encodings of primary texts as dependency quivers, and a different defensible encoding of Joslyn 1995 — one that does not make the control loop a cycle at quiver level — could dissolve the obstruction. Not a maximality result.”
2. **It is about Paths, which imposes no relations.** Impose one and it can die. If feedback is taken to be idempotent,  $\ell \circ \ell = \ell$ , then  $\text{Hom}(O_E, O_E)$  collapses to  $\{\text{id}, \ell\}$  and Theorem 1 fails. Whether  $\text{control}_2$  licenses that identification is a claim about your text, not about category theory.
3. **Re-bracketing does not dissolve it.** Your (31)  $\langle O_E, \langle C, O_I \rangle \rangle$  and (32)  $\langle O_I, \langle C, O_E \rangle \rangle$  regroup the vertices; neither removes an arrow, so the loop and the theorem survive both. (32) adds a second loop if the “kind of loop” of §3.7.2 is drawn, which only strengthens the obstruction.
4. **Unrolling does dissolve it.** Index the loop by level: vertices  $C, O_E^{(k)}, O_I^{(k)}$  for  $k \in \mathbb{N}$ ; arrows  $h: C \rightarrow O_E^{(0)}$ ,  $g_k: O_E^{(k)} \rightarrow O_I^{(k)}$ ,  $f_k: O_I^{(k)} \rightarrow O_E^{(k+1)}$ . This  $Q$  is acyclic, every hom-set is finite (at most one path between any two vertices), and faithful functors into  $\mathcal{I}_K$  are no longer excluded on cardinality grounds. *Stated, not machine-checked.*
5. **It does not say  $\text{control}_2$  is incomparable in principle.** Only that the method placing the six acyclic traditions cannot reach a cyclic one; a quotient, enrichment, or 2-categorical treatment of the loop is neither ruled out nor attempted.

### 4 The question, which only you can answer

In the margin of *Computational Systems Ontology: Towards Practical Integration of the Systems Sciences with BERT* (Thornton, August 2025, p. 4) you wrote: “One could argue that control is always so circular.” The encoding took that literally: one system, quiver level, no relations on the loop’s powers. Items 2 through 4 of §3 are then one question: *where does the closure of  $\text{control}_2$  live?* Your text already places one hierarchy: the second-order variation of Definition 28 “is at a hierarchically distinct level from the variation in  $O$  itself,” manifested “over a slower time scale or larger spatial scale” (p. 104), and  $O_E, O_I$  are “more tightly coupled to each other than either is to  $C$ ” (p. 107).  $Q_J$  reads that as: the hierarchy is  $C$  over  $O$  and is carried by  $h$  alone; the  $f, g$  loop is level-flat, so each  $\ell^n$  is a distinct trajectory within one level. The metasystem transition of §5.1 is a different event, a new level of control “emerging where one previously did not exist,” and you leave open whether it is  $\text{control}_1$  or  $\text{control}_2$ . The question is therefore: is that reading right, or does the “hierarchically distinct level” of p. 104 also index the loop, so that its passes unroll across levels and the cycle is an artefact of drawing every level on one page? Four outcomes, each a result:

1. **You defend the cycle.** Theorem 1 becomes author-endorsed.
2. **You propose an acyclic encoding** (the unrolling of §3 item 4, or another): the obstruction dissolves for it, and presentation-relativity is exhibited rather than asserted.
3. **You rule both encodings defensible.** One text, two shapes, different comparison behaviour: the sharpest exhibit yet for the validation layer no proof checker sees.
4. **None of these.** The quiver is the wrong instrument for the question, or the loop should be quotiented rather than drawn or unrolled (§3 item 2), or something not anticipated here. This is also a result, and likely the most informative one.

### 5 One provenance item, separately

Definition 28 is entry 009 of the definition atlas at **math.systems**, page-image transcribed and hand-checked, with five of your case rulings (thermostat and the decoupled-envelope pair admitted; damped spring, boiling liquid, spinning top refused). One grading is offered for correction: Definition 25 ( $\text{control}_1$ ) is introduced on p. 101 as Turchin’s 1992 definition, “we offer it again here,” so it is graded a restatement of Turchin and Definition 28 is recorded as the first control definition that is yours alone.

<sup>5</sup>`no_faithful_joslyn_to_klir`; the finiteness hypothesis is discharged by an instance built from `klirHomSubsingleton`, not by inspection in prose.